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Genome-Wide Population Structure in a Marine Keystone Species, the European Flat Oyster (*Ostrea edulis*)

Homère J. Alves Monteiro^{1,2}  | Dorte Bekkevold¹  | George Pacheco^{1,3} | Stein Mortensen⁴ | Runyang Nicolas Lou^{5,6}  | Nina O. Therkildsen⁵  | Arnaud Tanguy⁷ | Chloé Robert⁸  | Pierre De Wit^{8,9}  | Dorte Meldrup¹ | Ane T. Laugen^{10,11}  | Philine S. E. zu Ermgassen¹²  | Åsa Strand¹³  | Camille Saurel¹⁴  | Jakob Hemmer-Hansen¹ 

¹National Institute of Aquatic Resources, Technical University of Denmark, Silkeborg, Denmark | ²Section for Evolutionary Genomics, Faculty of Health and Medical Sciences, The Globe Institute, University of Copenhagen, Copenhagen K, Denmark | ³Department of Biosciences, Centre for Ecological and Evolutionary Synthesis, University of Oslo, Oslo, Norway | ⁴Institute of Marine Research, Bergen, Norway | ⁵Department of Natural Resources and the Environment, Cornell University, Ithaca, New York, USA | ⁶Department of Integrative Biology, University of California Berkeley, Berkeley, California, USA | ⁷Sorbonne Université, CNRS, UMR 7144, Station Biologique de Roscoff, Roscoff, France | ⁸Department of Marine Sciences, Tjärnö Marine Laboratory, University of Gothenburg, Strömstad, Sweden | ⁹Department of Biological and Environmental Sciences, University of Gothenburg, Gothenburg, Sweden | ¹⁰Department of Ecology, Swedish University of Agricultural Sciences, Uppsala, Sweden | ¹¹Department of Natural Sciences, Centre for Coastal Research, University of Agder, Kristiansand, Norway | ¹²Changing Oceans Group, School of Geosciences, University of Edinburgh, Edinburgh, UK | ¹³Department of Environmental Intelligence, IVL Swedish Environmental Research Institute, Fiskebäckskil, Sweden | ¹⁴National Institute of Aquatic Resources, Danish Shellfish Centre, Technical University of Denmark, Nykøbing Mors, Denmark

Correspondence: Homère J. Alves Monteiro (homerejalvesmonteiro@gmail.com) | Jakob Hemmer-Hansen (jhh@aqu.dtu.dk)

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ABSTRACT

Ostrea edulis, the European flat oyster, was once a widespread economically and ecologically important marine species, but has suffered dramatic declines over the past two centuries. Consequently, there has been a surge in European restoration efforts, many of which focus on restocking as a conservation measure. In this study, we used whole-genome sequencing (WGS) data to investigate the population structure, demographic history, and patterns of local adaptation of *O. edulis* across its natural distribution with increased sampling densities at Scandinavian localities. Results revealed seven distinct genetic clusters, including previously undescribed complex population structure in Norway, and evidence for introgression between genetic clusters in Scandinavia. We detected large structural variants (SVs) on three pseudo-chromosomes. These megabase long regions were characterised by strong linkage disequilibrium and clear geographical differentiation, suggestive of chromosomal inversions potentially associated with local adaptation. The results indicated that genomic traces of past translocations of non-native *O. edulis* were still present in some individuals, but overall, we found limited evidence of major impacts of translocations on the scale of contemporary population structure. Our findings highlight the importance of considering population structure and signatures of selection in the design of effective conservation strategies to preserve and restore wild native European flat oyster populations, and we provide direct knowledge safeguarding sustainable mitigation actions in this important species.

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